

Nakul Makol

Delhi, India

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Education

Thapar Institute of Engineering and Technology
Bachelor of Engineering in Electronics and Communication
CGPA-8.62

Aug. 2023 – June 2027
Patiala, Punjab

Relevant Coursework

- Artificial Intelligence
- Generative AI
- Agentic AI
- Database Management
- Data Structures
- Algorithm Analysis
- Computer Vision
- RAG/NLP

Experience

ZeroForg

June 2026 – August 2026

AI/ML Intern

Gurugram, Haryana

- Developed an end-to-end computer vision pipeline for automated detection and analysis of 16+ crack and surface defect categories using Python, OpenCV, and image processing techniques, improving industrial quality inspection efficiency.
- Implemented and optimized 10+ image processing algorithms including Sobel, Canny, Gabor, Otsu & Multi-Otsu Thresholding, Watershed, GLCM, and Contour Detection by tuning kernel sizes (3×3 – 9×9) and filter parameters to enhance edge detection while minimizing noise.
- Researched stereo vision-based depth estimation and evaluated YOLO, RF-DETR, DINOv2, and ResNet architectures to support scalable AI-driven defect detection systems.

Wibix Consulting

June 2025 – August 2025

Data Analyst Intern

Gurugram, Haryana

- Processed and validated 100K–150K records across 5+ internal and external sources, reducing data inconsistencies by 25–30%.
- Delivered insights from large-scale structured and unstructured data, uncovering actionable patterns that influenced 3+ business decisions.

Projects

Finsight AI – Intelligent Financial Copilot | *Flask/FastAPI, PostgreSQL, LLM, AI/ML, React, Railway*

- *Live Demo:* fin-sight-ai-weld.vercel.app
- Built an AI-powered financial analytics platform integrating portfolio management, real-time market insights, AI chat assistance, and news sentiment analysis, enabling users to monitor 50+ stocks with actionable investment recommendations.
- Integrated a responsive React + TypeScript dashboard featuring portfolio allocation, sector diversification, profit/loss analysis, watchlists, and financial visualizations, improving portfolio monitoring efficiency by 70%.
- Designed a scalable backend with Python, Flask/FastAPI, PostgreSQL, JWT authentication, and cloud deployment, implementing secure authentication, portfolio persistence, and optimized queries for 99%+ availability.

MediBot AI – Intelligent Medical Chatbot | *Flask, LangChain, FAISS, Hugging Face, Groq Llama 3.3, Railway*

- *Live Demo:* medical-chatbot-production-ed14.up.railway.app
- Built an AI-powered RAG medical chatbot using LangChain, FAISS, Hugging Face embeddings, and Groq Llama-3.3 70B, enabling context-aware responses from a semantic medical knowledge base with 95%+ retrieval relevance.
- Established a high-performance vector search pipeline generating embeddings with all-MiniLM-L6-v2 and indexing 1,000+ medical text chunks in FAISS, reducing retrieval time by ~45% while improving response relevance.
- Developed a responsive Flask-based full-stack web application with REST APIs, secure environment variable management, and Railway cloud deployment, achieving 99%+ application availability.

Technical Skills

Languages: Python, C++, C, HTML/CSS, JavaScript, SQL

AI/ML Computer Vision: OpenCV, Data Science, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, CNN, ResNet, YOLO, RF-DETR, DINOv2, Stereo Vision, Image Segmentation, Feature Extraction

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Postman, Docker, Linux, Power BI

Leadership / Extracurricular

AIIESEC

February 2024 – March 2025

Sales Head

TIET, Patiala

- Drove outreach campaigns engaging 200+ students, contributing to a 30% increase in new member sign-ups..
- Served on the organizing committee for the society fair, coordinating 5+ teams and managing event logistics for 1500+ attendees.
- Supported global exchange programs, assisting in the coordination of 20+ international exchange applications .